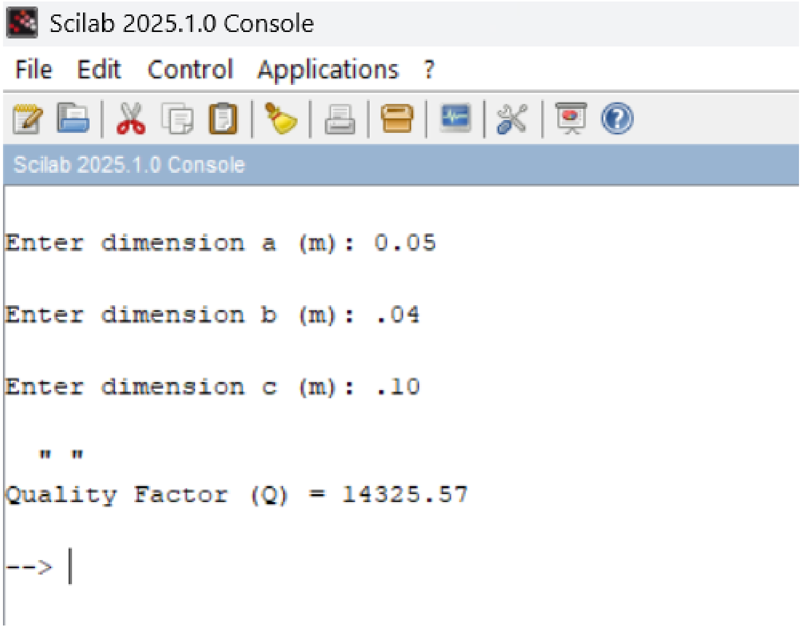
ECA15 TRANSMISSION LINE & WAVEGUIDES

EXP 6

Test Casse 1

```
clc;
clear;
close;

// Constants
sigma = 5.8e7;      // Conductivity of copper (S/m)
mu0 = 4 * %pi * 1e-7;
f101 = 3.335e9;     // Resonant frequency (Hz)

// Dimensions (meters)
a = input("Enter dimension a (m): ");
b = input("Enter dimension b (m): ");
c = input("Enter dimension c (m): ");

// Skin depth
d = sqrt(1 / (%pi * f101 * mu0 * sigma));

// Q-factor for TE101 mode
Q = ((a^2 + c^2) * a * b * c) / ...
    (d * (2 * b * (a^3 + c^3) + a * c * (a^2 + c^2)));

disp(" ");
printf("Quality Factor (Q) = %.2f\n", Q);
```

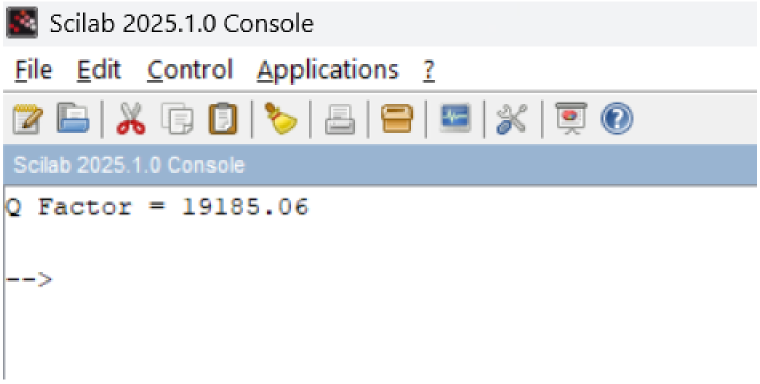
Output:



Test case 2:

```
clc;
clear;

sigma = 5.8e7;
mu0 = 4*%pi*1e-7;
f101 = 3.335e9;

a = 0.07;
b = 0.05;
c = 0.12;

d = sqrt(1/(%pi*f101*mu0*sigma));

Q = ((a^2+c^2)*a*b*c)/(d*(2*b*(a^3+c^3)+a*c*(a^2+c^2)));

printf("Q Factor = %.2f\n",Q);
```

Output


Scilab 2025.1.0 Console
File Edit Control Applications ?
Scilab 2025.1.0 Console
Q Factor = 19185.06
-->